

Srushti Nerkar

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EDUCATION

- **Malaviya National Institute of Technology, Jaipur** 2026
Computer Science and Engineering
- **Unnati High School** 2022
Higher Secondary School Certificate, Maharashtra

TECHNICAL SKILLS AND INTERESTS

Languages: Java, Python, C++, JavaScript, SQL

Generative AI & Agentic Systems : RAG (vector retrieval, embeddings), LLM APIs, Tool-Calling / Agent Loops, Prompt Engineering, PyTorch, Scikit-learn

Databases & Caching: PostgreSQL (pgvector), MySQL, MongoDB, Redis

System Design & Architecture: HLD (Microservices, Scalability), LLD (Design Patterns, SOLID Principles)

Backend & Frameworks: Spring Boot, Node.js, Express.js

DevOps & Tools: Docker, Git, GitHub, VS Code, Linux/Unix (Shell Scripting)

Networking & OS: TCP/IP, Linux Networking Commands

EXPERIENCE

- **Birdvision Consulting Pvt. Ltd.** May 2025 - July 2025
Software Development Intern Pune
 - Designed RESTful microservices powering a conversational WhatsApp Business interface for 10K+ users, handling message routing and intent-based status updates across distributed services.
 - Built services for messaging, routing, and status handling, applying separation-of-concerns and clean service boundaries.
 - Improved system reliability by optimizing API error handling and response processing across distributed endpoints.

PROJECTS

- **AeroPath: Intelligent Connected Travel Timeline Engine | GitHub** Ongoing
React 19, TypeScript, Node.js, Express.js, PostgreSQL (pgvector), Prisma ORM, Firebase Auth, Java, Spring Boot, Docker, LLM APIs
 - Architected a full-stack travel platform integrating real third-party data into search flows, querying live flight schedules via Aero-DataBox API and hotel listings via Hotels4 API to replace mock data with backend-driven results.
 - Engineered a secure user synchronization pipeline bridging Firebase Client Auth with a Node.js backend using Firebase Admin SDK and Prisma ORM to map verified user records atomically into PostgreSQL.
 - Developed a Spring Boot microservice using Spring Data JPA and Docker for disruption alert logic, transitioning the system into a polyglot distributed architecture.
 - Building a conversational assistant combining LLM tool-calling across existing flight, hotel, and geocoding APIs with RAG retrieval over a pgvector-backed travel knowledge base, including source citations.
- **PRISM: Progressive Routing for Intrusion Detection via Spiking Multi-layer Networks** May 2026
Python, PyTorch, Scikit-learn, Spiking Neural Networks (SNN), Isolation Forest, Ensemble Learning
 - Co-engineered PRISM, a hierarchical IDS cascading an Isolation Forest global pre-filter with a 3-layer Parametric LIF Spiking Neural Network (SNN) and a 4-model stacked ensemble.
 - Resolved threshold contradictions in hierarchical IDS by decoupling anomaly detection from classification, intercepting zero-day threats via feature-space signatures before misrouting by the binary SNN classifier.
 - Achieved 95.59% overall accuracy and a 0.758 macro F1-score on 377,698 test flows (CICIDS2017), securing a 60.3% overall zero-day detection rate at a low 3.39% false positive rate.
- **Smart Sentry: Real-Time Industrial Anomaly Detection Pipeline | GitHub** Oct 2025
Python, Scikit-learn, Pandas, Kafka, Redis, PostgreSQL, Streamlit, Docker Baker Hughes Hackathon
 - Engineered a phase-aware anomaly detection system using Kafka and Redis to minimize false positives during transient operational states (startup/shutdown).
 - Achieved 98% overall accuracy and a 0.71 F1-score on the anomaly class, isolating critical failures under peak data load.

CODING PROFILE

- **LeetCode (Max Rating - 2025, Top 2.3%)** Global Rank 256 in Biweekly Contest 190 | Link
- **Geeks for Geeks (Max Rating - 1756, 3 star)** Global Rank 210 in Weekly 189 | Link
- **CodeChef (Max Rating - 1652, 3 star)** Global Rank 194 in Starters 257 | Link

ACHIEVEMENTS

- **Finalist at Baker Hughes Hackathon 2025** 2025
- **Granted 4 Indian patents for technological innovations**
- **Co-authored "PRISM: Progressive Routing for Intrusion Detection via Spiking Multi-layer Networks"** 2026
submitted to ICMLDE 2026